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REVIEW

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Quantifying demands on the hamstrings during high-speed running: A systematic review and meta-analysis

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Abstract

Introduction: Hamstring strain injury (HSI) remains a performance, economic, and player availability burden in sport. High-speed running (HSR) is cited as a common mechanism for HSI. While evidence exists regarding the high physical demands on the hamstring muscles in HSR, meta-analytical synthesis of related activation and kinetic variables is lacking.

Methods: A systematic search of Medline, Embase, Scopus, CINAHL, SportDiscus, and Cochrane library databases was conducted in accordance with the PRISMA 2020 guidelines. Studies reporting hamstring activation (electromyographic [EMG]) or hamstring muscle/related joint kinetics were included where healthy adult participants ran at or beyond 60% of maximum speed (activation studies) or 4 m per second (m/s) (kinetic studies).

Results: A total of 96 studies met the inclusion criteria. Run intensities were categorized as "slow," "moderate," or "fast" in both activation and kinetic based studies with appropriate relative, and raw measures, respectively. Meta-analysis revealed pooled mean lateral hamstring muscle activation levels of 108.1% (95% CI: 84.4%–131.7%) of maximal voluntary isometric contraction (MVIC) during "fast" running. Meta-analysis found swing phase peak knee flexion internal moment and power at 2.2 Newton meters/kilogram (Nm/kg) (95% CI: 1.9–2.5) and 40.3 Watts/kilogram (W/kg) (95% CI: 31.4–49.2), respectively. Hip extension peak moment and power was estimated as 4.8 Nm/kg (95% CI: 3.9–5.7) and 33.1 W/kg (95% CI: 17.4–48.9), respectively.

Conclusions: As run intensity/speed increases, so do the activation and kinetic demands on the hamstrings. The presented data will enable clinicians to incorporate more objective measures into the design of injury prevention and return-to-play decision-making strategies.

KEYWORDS

activation, hamstring, kinetics, sprinting

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1 | INTRODUCTION

Hamstring strain injury (HSI) is a major performance, economic, and availability burden in many sports, producing the greatest time loss in Australian football,[1] up to 15% of all injuries in rugby[2] and 19% of all injuries in professional football.[3] Despite the recent attention on HSI risk reduction strategies, the cost of HSI continues to climb, largely due to injuries doubling between 2001/2002 and 2021/2022,[3] combined with rising athlete salaries.[4] The burden of HSI extends to athletic performance through its impacts on training and game availability, which is linked to sporting success in both team[5] and Olympic sports.[6] Despite the growing knowledge on risk factors[7] recurrence rates for HSI are high and subsequent injuries often leave the athlete with greater impairments and longer times to return to sport.[8] High prevalence of subsequent HSI suggests that current rehabilitation practices are not sufficient to minimize the burden of HSI and cannot prevent or predict re-injury.[9] Novel, effective methods to reduce injury rates remain of high interest in the sport medicine community.

The mechanism of injury in 57%–78% of HSI cases in field-based sports is cited as high-speed running (HSR).[10,11] During late swing in HSR, hamstring muscle kinetics,[12] activation and strain reach a peak,[13] indicating high demands and a potential for injury within this phase of the gait cycle. However, evidence exists that kinetic demands (i.e., hip and knee joint moments) also peak during early stance,[14,15] highlighting another "high-demand" phase of the gait cycle where injury may occur.[16] Despite some studies showing a relationship between specific running mechanical variables and HSI,[17,18] current evidence of causation is limited.[19]

Lack of knowledge for a sports demands can lead to athletes being underprepared for their sport, influencing subsequent injury risk.[20] A better understanding of hamstring demands during specific sporting tasks associated with injury may facilitate the development of targeted prevention and rehabilitation strategies for HSI. With knowledge regarding hamstring muscle activation, muscle and related joint kinetics, clinicians can match known exercise demands in a controlled rehabilitation setting before exposure to various intensities in return to running programs. This paper seeks to systematically review the evidence quantifying the activation and kinetic demands on the hamstrings in healthy adults during HSR.

2 | MATERIALS AND METHODS

2.1 | Protocol

This systematic review was written in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.[21] The review protocol was registered on 01/07/2022 with PROSPERO International Prospective Register for Systematic Reviews website (CRD 42022339074).

2.2 | Literature search strategy

A comprehensive strategy (Appendix S1) was devised and implemented on 06/12/22 to search databases; Medline, Embase, Cochrane CENTRAL, CINAHL, SportDiscus, and Scopus. Key words and database medical subject headings were used in three search themes: (1) hamstring muscles and the related planes of action at both the hip and knee joints; (2) terms relating to quantifying kinetics and/or muscle activation; and (3) high-speed running. Database search was limited to publications in English, with no date or publication type restrictions.

Two authors (TM and SB) independently reviewed title and abstracts for eligibility using online systematic review software (Covidence systematic review software, Veritas Health Innovation, Melbourne, Australia). Full-text articles were included in the review if they met the selection criteria. Conflicts where no consensus could be made were resolved by discussion with a third reviewer. Reference lists of included articles were searched for further eligible publications.

2.3 | Selection criteria

Observational or intervention-based studies were included where healthy adult participants (18 years and over) performed, steady-state, high-speed, straight line running on a firm surface in a shod footwear condition. The threshold for the experimental task to be considered "high-speed running" has been reported as low as 4 m/s in a recent review defining sprinting in teams sport.[22] The same review cites "sprinting" velocity in team sport having a minimum threshold of 7 m/s.[22] High-speed running was therefore defined as running at or beyond 60% of maximum speed (muscle activation studies) or 4 m per second (m/s) (kinetic studies) and has been used as the minimum criteria for inclusion in this review. Studies where experimental tasks of high-speed running focused on acceleration, deceleration, curvilinear, and change of direction running data were excluded, due to the associated deviation in mechanics compared with straight line running.[23,24] Overground running studies were limited to flat surfaces, while treadmill-based studies were included on two conditions; (1) treadmill gradient was set between 0% and 1% and (2) gravity-altering treadmills were not used to alter subject's normal mass

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due to their impact on the outcomes measured.[25,26] If participants' data in a study were recorded only in fatigued states, and/or if exact run speed/relative intensity during data collection was not stated, the study was excluded. If data were presented in an ambiguous manner (i.e., unclear graph or numerical data presentation), authors were contacted and if no response was received, the publication were excluded.

Outcomes of interest and their specific selection criteria could be divided into part A: studies reporting hamstring muscle activation (via electromyograph [EMG] or muscle function magnetic resonance imaging); and part B: studies reporting muscle kinetics or joint kinetics (relevant hip or knee joint planes).

2.3.1 | Part A: muscle activation

EMG studies were included in this review when EMG magnitudes were normalized by levels obtained during maximal voluntary isometric contraction (MVIC) or during maximal sprinting. As muscle activation is a relative measure, no raw speed threshold was set, but publications reporting EMG data only at running intensities below 60% of maximal speed were excluded. Spatial activation measures of hamstring muscles captured via muscle function magnetic resonance imaging were also considered for inclusion.

2.3.2 | Part B: joint kinetics

Estimates of hamstring muscle force, internal joint moment, rate of moment development, joint power, joint work, and joint impulse were included for the primary outcomes of hip extension and knee flexion; and then for the secondary outcomes of hip adduction, hip internal rotation (IR), hip external rotation (ER), knee IR, knee ER, knee adduction, and knee abduction. Joint work and power data were not extracted when direction (and therefore relevance to hamstrings) could not be determined.

2.4 | Data extraction

Participant characteristics (age, sex, height, body mass, sport, and level of participation), sample size, experimental task (running speed and/or percentage of maximal sprint speed), data collection (equipment, sampling rates), and analysis were extracted from all included studies into a Microsoft Excel spreadsheet by one reviewer (TM) and confirmed by another (SB). Where relevant data were presented only in graphical form, the online software WebPlotDigitiser (accessed at https://automeris.io/WebPlotDigitizer/) was utilized to extract numerical data points. One reviewer (TM) then extracted data for the review's subsections and a second confirmed (SB), as follows. Unless otherwise specified, both activation and kinetic demands were quantified as peak magnitudes (scalar quantities) from any phase of the gait cycle.

2.4.1 | Part A: muscle activation

Normalized hamstring EMG data were extracted and clustered into relative running intensities: 60.0%–74.9%, 75.0%–89.9%, and 90.0%–100.0% (of maximal sprint speed). No muscle function magnetic resonance imaging studies met the inclusion criteria and thus none were included in the review.

2.4.2 | Part B: joint kinetics

Individual hamstring muscle force, muscle power, muscle work, net internal joint moment, rate of moment development, joint work, joint power, and joint impulse were extracted into a Microsoft Excel spreadsheet by one reviewer (TM) and confirmed by another (SB). Where publications provided kinetic metrics but incomplete run speed data (i.e., on sub-maximal trials), maximal run speed and relative running intensity measures (i.e., percentage of maximum) were used to manually calculate absolute run speed. Hamstring muscles may contribute force only to the planes of motion their lever arm permits,[27,28] therefore, kinetic reporting was limited to hip extension, hip and knee adduction, knee flexion, knee abduction, hip and knee IR and ER.

2.5 | Data synthesis

To be included in meta-analysis, studies had to provide sufficient statistical details (e.g., means, standard deviations), score a minimum of "moderate" on their respective methodological quality scales and utilize appropriate/homogenous normalization (e.g., Nm/kg for kinetics or percentage MVIC for EMG). Subgrouping of running speeds for meta-analysis of kinetic metrics was informed by data extraction: (1) 4.00–6.49 m/s (slow), (2) 6.50–8.49 m/s (moderate), and (3) 8.50+ m/s (fast). Pooled means with 95% confidence intervals were presented using Comprehensive Meta-analysis software V.XX (Biostat, Englewood, New Jersey, USA). Heterogeneity was measured with prediction intervals and I^2 tests using the DerSimonian and Laird (DL) method.

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2.6 | Methodological quality

Assessment of methodological quality was carried out independently by two reviewers (TM and SB) using the Joanna Briggs Institute (JBI) Critical Appraisal Checklist for Analytical Cross-Sectional Studies.[29] When agreement could not be reached by discussion, a third reviewer was consulted. The JBI tool was selected as it is recommended and relevant to most studies included in the review.[30] The tool includes eight questions tailored to cross-sectional studies where reviewers answer "Yes," "No," or "Unclear," Questions 4 and 8 were marked "N/A" due to their lack of relevance to the types of included studies, leaving a maximum total of six possible points. Articles were scored by reviewers according to the response to each question, awarding 1 point for "Yes" and 0 points for a "No" or "Unclear." Scores were tallied for each included article and summarized (in Tables 1 and 2). As recommended by the JBI reviewers' manual[31] an agreement between reviewers regarding cut-offs was reached via discussion: (i) low risk of bias if studies reached ≥70% "yes" scores; (ii) moderate risk of bias if "yes" scores were 50%–69%; and (iii) high risk of bias if "yes" scores were ≤49%. Study quality was downgraded by one category where appropriate (i.e., moved from low to moderate risk of bias, or moderate to high risk of bias) if the criterion for Question 7 of the checklist was not met due to its importance for the validity of the measures.

Due to the methodological difficulties associated with EMG analysis, a checklist specific to methodological quality published by Richardson and colleagues[32] was also implemented (Table 3). The quality assessment tool has a total possible 13 points, but modified scores out of 11 were extracted here due to two questions not being relevant to the studies included in this review. Total scores >80% were considered low risk, between 70% and 80% moderate risk, and <70% high risk of bias.

3 | RESULTS

3.1 | Search, study selection, and characteristics

The database search retrieved a total of 21 015 articles, from which 10 796 duplicates were removed. Title and abstract screening found 9831 articles to be irrelevant, leaving 388 titles for full-text review. Applying the inclusion/exclusion criteria, 301 titles were excluded leaving 88 eligible publications in this review. Reference checking of the included articles found a further 9 eligible studies, which were added for data extraction. After exclusion of

TABLE 1 Methodological quality: JBI checklist for analytical cross-sectional studies (EMG).

Study	Inclusion criteria	Subjects and setting described in detail	Exposure measures valid and reliable	Objective standard criteria for condition	Confounding factors identified	Strategies for confounding factors stated	Outcomes measures valid and reliable	Appropriate statistical analysis	Total /6
Belli et al. 2002[33]	0	1	1	N/A	0	0	1	N/A	3
Hegyi et al. 2019[40]	1	1	1	N/A	1	0	1	N/A	5
Higashihara et al. 2010[41]	1	0	1	N/A	0	0	1	N/A	3
Jonhagen et al. 1996[42]	1	0	1	N/A	1	0	1	N/A	4
Kakehata et al. 2021[34]	0	1	1	N/A	1	0	1	N/A	4
Kuitunen et al. 2002[35]	0	1	1	N/A	1	1	0	N/A	4
Kyrolainen et al. 1999[38]	0	1	1	N/A	1	0	0	N/A	3
Kyrolainen et al. 2001[37,123]	1	1	1	N/A	1	0	0	N/A	4
Kyrolainen et al. 2005[36]	0	1	1	N/A	1	0	0	N/A	3
Ono et al. 2015[15]	1	1	1	N/A	0	0	0	N/A	3
Pietraszewski et al. 2021[39]	1	1	1	N/A	1	0	1	N/A	5
Verheul et al. 2017[43]	1	1	1	N/A	0	0	1	N/A	4

Note: High risk of bias = 🟥, moderate risk of bias = 🟧, low risk of bias = 🟩.

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TABLE 2 Methodological quality: JBI checklist for analytical cross-sectional studies (Kinetics).

Study	Inclusion criteria clearly defined	Subjects and setting described in detail	Exposure measures valid and reliable	Objective standard criteria for condition	Confounding factors identified	Strategies for confounding factors stated	Outcomes measures valid and reliable	Appropriate statistical analysis	Total /6
Almonroeder et al. 2015[45]	1	1	1	N/A	1	1	1	N/A	6
Almonroeder et al. 2016[44]	1	1	1	N/A	0	0	1	N/A	4
Alt et al. 2021a[79,80]	1	1	1	N/A	1	1	1	N/A	6
Alt et al. 2021b[79,80]	1	1	1	N/A	1	1	1	N/A	6
Arampatzis et al. 1999[81]	0	0	1	N/A	0	0	0	N/A	1
Arnold et al. 2013[46,140]	1	1	0	N/A	0	0	0	N/A	2
Belli et al. 2002[33]	0	1	1	N/A	0	0	0	N/A	2
Bezodis et al. 2008[82]	0	0	1	N/A	1	0	1	N/A	3
Bonacci et al. 2013[83]	1	1	1	N/A	0	0	1	N/A	4
Borgia et al. 2021[47]	1	1	1	N/A	0	0	1	N/A	4
Buczek et al. 1990[84]	1	1	1	N/A	1	1	0	N/A	5
Chapman et al. 1984[85]	0	0	1	N/A	0	0	0	N/A	1
Chumanov et al. 2011[72]	1	1	1	N/A	0	0	1	N/A	4
Creaby et al. 2017[48]	1	1	1	N/A	0	0	0	N/A	3
deDavid 2015[86]	1	1	1	N/A	1	0	1	N/A	5
DeVita et al. 1990[87]	0	0	1	N/A	0	0	0	N/A	1
Diss et al. 2015[113]	1	1	1	N/A	0	0	1	N/A	4
Donnelly et al. 2012[120]	1	1	1	N/A	0	0	1	N/A	4
Dorn et al. 2012[73]	1	1	1	N/A	1	0	1	N/A	5
Dunn et al. 2018[122]	1	1	1	N/A	0	0	1	N/A	4
Ema et al. 2018[114]	1	1	1	N/A	1	1	1	N/A	6
Flores et al. 2021[49]	1	1	1	N/A	0	0	1	N/A	4
Fukuchi et al. 2015[50]	1	0	1	N/A	0	0	1	N/A	3
Gehring et al. 2014[51]	1	0	1	N/A	1	1	1	N/A	5
Glitsch et al. 1997[88]	0	0	1	N/A	1	1	0	N/A	3
Graf et al. 2017[52]	1	1	1	N/A	0	0	1	N/A	4
Haneda et al. 2003[71]	1	0	1	N/A	1	1	1	N/A	5
Harrison et al. 1986[74]	1	1	1	N/A	0	0	0	N/A	3
Higashihara et al. 2018[23]	1	1	1	N/A	1	0	1	N/A	5
Hirosuke et al. 2013[115]	0	1	1	N/A	0	0	0	N/A	2

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TABLE 2 (Continued)

Study	Inclusion criteria clearly defined	Subjects and setting described in detail	Exposure measures valid and reliable	Objective standard criteria for condition	Confounding factors identified	Strategies for confounding factors stated	Outcomes measures valid and reliable	Appropriate statistical analysis	Total /6
Howe et al. 2015[89]	1	1	1	N/A	0	0	1	N/A	4
Huang et al. 2013[90]	1	1	1	N/A	1	1	1	N/A	6
Hunter et al. 2004[91]	0	1	1	N/A	0	0	1	N/A	3
Kenneally-Dabrowski et al. 2019[18]	1	1	1	N/A	1	1	1	N/A	6
Khassetarash et al. 2020[92]	1	1	1	N/A	0	0	1	N/A	4
Krabbe et al. 1997[93]	1	0	1	N/A	0	0	1	N/A	3
Kosonen et al. 2017[53]	1	0	1	N/A	0	0	1	N/A	3
Kuhman et al. 2016[94]	1	0	1	N/A	0	0	1	N/A	3
Kuitunen et al. 2002[35]	0	1	1	N/A	1	1	0	N/A	4
Kulmala et al. 2013[54]	1	0	1	N/A	1	1	1	N/A	5
Kulmala et al. 2014[95]	1	0	1	N/A	0	0	1	N/A	3
Kulmala et al. 2017[53]	1	1	1	N/A	0	0	1	N/A	4
Kyrolainen 2001[123]	1	1	1	N/A	1	0	0	N/A	4
Lewinson et al. 2013[56]	1	0	1	N/A	0	0	1	N/A	3
Lewinson et al. 2019[57]	1	1	1	N/A	0	0	1	N/A	4
Mann et al. 1980[141]	0	0	1	N/A	0	0	0	N/A	1
Mitchell et al. 2022[97]	1	1	1	N/A	0	0	1	N/A	4
Moffit et al. 2020[98]	1	0	1	N/A	1	1	1	N/A	5
Muendermann et al. 2003[58]	0	1	1	N/A	1	0	1	N/A	4
Nagahara et al. 2017a[99]	0	1	1	N/A	1	1	0	N/A	4
Nagahara et al. 2017b[100]	1	1	1	N/A	1	1	1	N/A	6
Nagano et al. 2014[142]	1	1	1	N/A	1	1	0	N/A	5
Nigg et al. 2003[59]	1	0	1	N/A	0	0	1	N/A	3
O'Meara et al. 2012[116]	0	1	1	N/A	1	1	0	N/A	4
Petersen et al. 2014[101]	1	1	1	N/A	1	0	0	N/A	4
Radzak et al. 2017[60]	1	1	1	N/A	0	0	0	N/A	3
Radzak et al. 2020[61]	1	1	1	N/A	0	0	0	N/A	3
Richert et al. 2019[62]	1	1	1	N/A	1	1	1	N/A	6

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TABLE 2 (Continued)

Study	Inclusion criteria clearly defined	Subjects and setting described in detail	Exposure measures valid and reliable	Objective standard criteria for condition	Confounding factors identified	Strategies for confounding factors stated	Outcomes measures valid and reliable	Appropriate statistical analysis	Total /6
Rooney et al. 2013[75]	1	1	1	N/A	0	0	1	N/A	4
Sado et al. 2016[138]	0	1	1	N/A	1	1	1	N/A	5
Sado et al. 2017[117]	0	1	1	N/A	1	1	1	N/A	5
Sakaguchi et al. 2015[63]	1	1	1	N/A	0	0	1	N/A	4
Sanno et al. 2018[118]	1	1	1	N/A	0	0	1	N/A	4
Schache 2009[103]	1	0	1	N/A	1	1	0	N/A	4
Schache et al. 2010[76]	1	0	1	N/A	1	1	0	N/A	4
Schache et al. 2011[102]	1	1	1	N/A	1	0	1	N/A	5
Schache et al. 2012[12]	1	1	1	N/A	1	0	1	N/A	5
Simpson et al. 1990[104]	1	0	1	N/A	0	0	1	N/A	3
Sinclair et al. 2015[64]	1	1	1	N/A	1	1	0	N/A	5
Sinclair et al. 2017[66]	1	1	1	N/A	0	0	1	N/A	4
Sinclair et al. 2021[65]	1	1	1	N/A	0	0	1	N/A	4
Sprague et al. 1983[105]	1	0	1	N/A	1	0	0	N/A	3
Stearne et al. 2014[111]	1	1	1	N/A	1	1	1	N/A	6
Stefanyshyn et al. 2006a[67]	1	1	1	N/A	0	0	1	N/A	4
Stefanyshyn et al. 2006b[67]	1	1	1	N/A	0	0	1	N/A	4
Stickley et al. 2018[68]	1	1	1	N/A	0	0	0	N/A	3
Stief et al. 2008[69]	1	1	1	N/A	0	0	1	N/A	4
Stoffel et al. 2010[121]	1	1	1	N/A	0	0	1	N/A	4
Sun et al. 2015[106]	1	1	1	N/A	1	1	1	N/A	6
Swanson et al. 2000[107]	1	0	1	N/A	0	0	1	N/A	3
Vardaxis 1990[108]	0	1	1	N/A	1	1	0	N/A	4
Willer et al. 2021[109]	1	1	1	N/A	0	0	1	N/A	4
Yi 2015[110]	0	0	0	N/A	0	0	0	N/A	0
Yokozawa et al. 2007[77]	0	0	1	N/A	0	0	0	N/A	1
Yu et al. 2016[119]	1	1	1	N/A	1	1	1	N/A	6
Zhang et al. 2021[70]	1	1	1	N/A	1	1	1	N/A	6
Zhong et al. 2017[112]	1	1	1	N/A	1	1	1	N/A	6

Note: High risk of bias = 0, moderate risk of bias = 0, low risk of bias = 0.

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TABLE 3 EMG methodological quality (from Richardson et al. 2020).

Study	Electrode type	Recording equipment	Sample rate	Skin preparation	Electrode placement	Raw signal	Lost data	Smoothing	Normalization	Timing of EMG	Test procedures	Fatigue	Statistical analyses	Total /11
Belli et al. 2002[33]	1	1	1	0	1	0	0	0	1	N/A	1	1	N/A	7
Hegyi et al. 2019[40]	1	0	1	1	1	1	1	1	1	N/A	1	1	N/A	10
Higashihara et al. 2010[41]	1	0	1	1	1	0	0	1	1	N/A	1	1	N/A	8
Jonhagen et al. 1996[42]	1	1	1	1	1	1	1	1	1	N/A	1	1	N/A	10
Kakehata et al. 2021[34]	1	0	1	1	1	1	0	1	1	N/A	1	1	N/A	9
Kuitunan et al. 2002[35]	1	1	1	0	0	1	1	1	1	N/A	1	0	N/A	8
Kyrolainen et al. 1999[38]	1	1	1	0	1	1	0	1	0	N/A	1	1	N/A	8
Kyrolainen et al. 2001[37]	1	1	1	0	1	1	0	0	0	N/A	1	1	N/A	7
Kyrolainen et al. 2005[36]	1	1	1	0	1	1	0	0	0	N/A	1	1	N/A	7
Ono et al. 2015[15]	1	0	1	1	1	0	1	1	1	N/A	1	1	N/A	9
Pietraszewski et al. 2021[39]	1	1	1	1	1	0	0	1	1	N/A	1	1	N/A	9
Verheul et al. 2017[43]	1	1	1	1	1	0	0	1	0	N/A	1	0	N/A	7

Note: High risk of bias = 0, moderate risk of bias = 0, low risk of bias = 0.

one paper due to unclear data presentation, the review included a total of 96 publications. Several papers reported both Part A kinetic and Part B EMG data, leading to kinetic data extracted from 86 studies, while EMG data were extracted from 12 studies. Study characteristics are fully detailed in Appendix S2 and a PRISMA flowchart is in shown in Figure 1.

3.2 | Methodological quality

The methodological quality findings for the JBI checklist (activation papers), JBI checklist (kinetic papers) and the EMG specific checklist are presented in Tables 1–3, respectively. The average score of activation studies from the modified JBI checklist was 62.5%, with two studies considered at low risk of bias, five at moderate risk, and five at high risk. The kinetic paper average score was 66.1%, with 23 studies considered at low risk, 41 at moderate risk, and 24 at high risk of bias. The average score of activation studies from the EMG checklist was 75%, with five studies considered at low risk of bias, 3 at moderate risk, and four at high risk.

3.3 | Electromyographic studies

We identified 12 studies assessing EMG activation of either lateral hamstrings ($n=7$)[33–39] or both medial and lateral hamstrings ($n=5$)[15,40–43] during running at 60% of maximum or higher. EMG data were normalized through one of three mechanisms: MVIC ($n=7$),[15,34,36,39–42] maximal sprint speed ($n=1$)[43] or manually normalized to maximal sprint speed by author (TM) ($n=4$).[33,35,37,38]

Relative to MVIC, maximal sprinting elicited 106.9% activation of medial hamstrings[42] in the only relevant study, while lateral hamstring activation ranged from 81.6%[42] to 145.3%[34] of MVIC. Only the lateral hamstrings activation during "fast" (90%–100% maximum) running was eligible for meta-analysis, where the pooled mean of 4 studies was 108.1% (95% CI: 84.4–131.7) (see Figure 2). Studies presenting submaximal running intensities reported heterogeneous EMG data. Incremental run speeds and their corresponding EMG magnitude ranges are described in Table 4 and results from all included papers are detailed in Appendix S3.

3.4 | Kinetics studies

Kinetic metrics relevant to the hamstrings were identified in 88 studies, 10 reporting on hamstring muscle-based kinetics, and 83 on joint-based kinetics to which the

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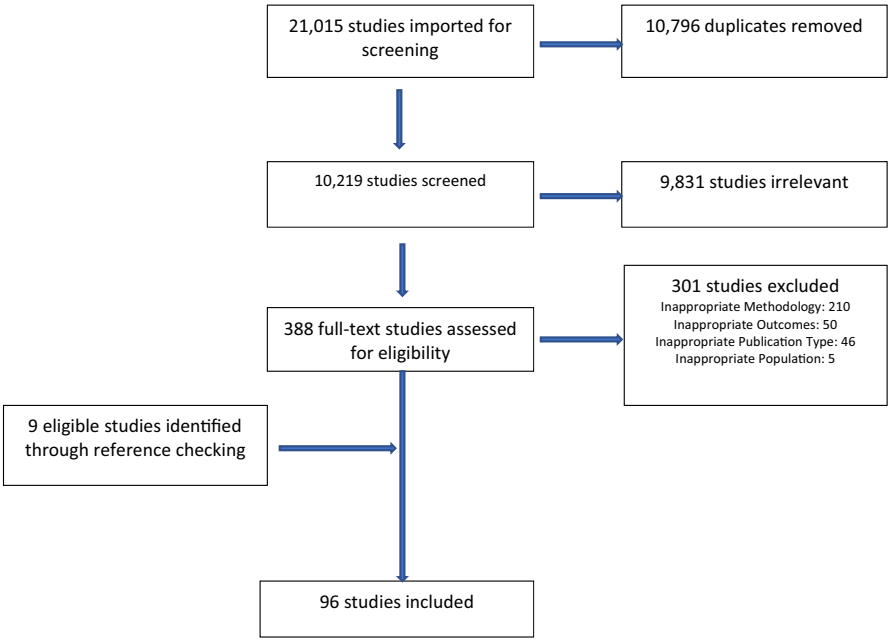
FIGURE 1 PRISMA flowchart.



Study name	Statistics for each study						
	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Higashihara 2010	91.95	4.54	20.64	83.05	100.85	20.24	0.00
Jonhagen 1996	81.59	5.46	29.79	70.89	92.29	14.95	0.00
Kakehata 2021	145.27	21.10	445.31	103.91	186.63	6.88	0.00
Pietraszewski 2021	132.00	8.14	66.27	116.04	147.96	16.21	0.00
Pooled	108.05	12.06	145.49	84.41	131.69	8.96	0.00
Prediction Interval	108.05			0.46	215.64		

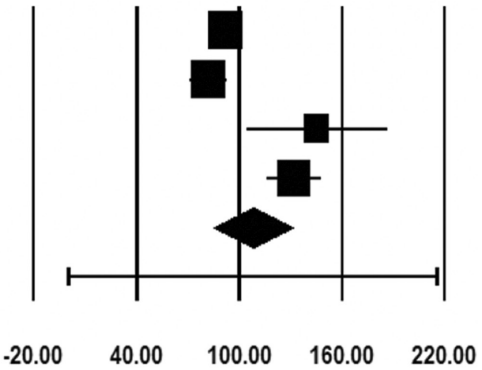


FIGURE 2 Forest plot of activation (EMG) magnitude (% maximum) for lateral hamstrings during "fast" running (90%–100% maximal intensity). Random effects model, $N=4$, heterogeneity: $I^2=90.8$.[34,39,41,42] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

hamstrings may contribute. Run speeds captured ranged from 4.0 m/s in many included studies ($n=30$),[33,36,44–70] up to 10.0 ± 0.3 m/s.[71]

Muscle kinetics data were extracted for semitendinosus (ST) ($n=5$),[12,46,64,65,72] semimembranosus (SM) ($n=4$),[12,64,65,72] biceps femoris (BF) ($n=1$),[72] biceps femoris short head (BFsh) ($n=3$),[12,64,65] biceps femoris long head (BFlh) ($n=4$),[12,46,64,65] and the collective hamstring muscle group (HAMS) ($n=5$).[73–77] The metrics extracted were muscle force ($n=8$),[12,46,64,65,72,73,76,77] muscle power ($n=2$),[12,76] muscle work ($n=4$),[12,72,73,76] percentage of muscle simulated maximum ($n=1$),[46] and contribution to hip extension/knee flexion internal moment ($n=1$)[77] (Appendix S4). Meta-analyses were not performed for muscle kinetics due to heterogeneity in methods of calculation and unit presentation. The maximum individual muscle force reported during sprinting was highest in SM at 46.8 ± 6.3 N/kg[12] (8.95 m/s), while 8.95 ± 1.66 BW was reported for the collective HAMS[73] (8.99 m/s). Muscle power during sprinting was reported as either positive (generation/concentric muscle action) or negative (absorption/eccentric muscle action), and values peaked in SM at 7.7 ± 1.1 and -22.4 ± 5.4 W/kg[12] (8.95 m/s), while HAMS peaked 15.0 ± 1.6 W/kg and -24.2 W/kg[76] (7.44 m/s). Muscle work during sprinting also was divided into positive (concentric) and negative (eccentric) work, and magnitudes peaked for SM: 0.47 ± 0.08 and -0.69 ± 0.10 J/kg[72] (7.8 m/s) while and HAMS peaked at -1.75 ± 0.31 J/kg[73]

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TABLE 4 Activation (EMG) range according to run speed.

Run intensity	Normalized to % maximal sprint		Normalized to % MVIC	
	Medial hamstrings peak activation (range)	Lateral hamstrings peak activation (range)	Medial hamstrings peak activation (range)	Lateral hamstrings peak activation (range)
60%–69.9%	-	38.4%–75.0% 37, 38	96.1% 40	38.6%–86.7% 36, 40
70%–79.9%	33.2% 44	37.3%–84.4% 38, 44	48.8%–113.3% 40, 41	49.3%–107.8% 40, 41
80%–89.9%	-	78.3%–90.3% 35, 38	76.1% 41	68.2%–72.2% 36, 41
90%–100%	100% 44	100% 33, 35, 37, 38, 44	34.6%–106.9% 42, 43	42.8%–145.3% 34, 43

(8.99 m/s). Although kinetic metrics peaked in muscles other than BFlh, the clinical relevance of this muscle as the most frequently injured hamstring[78] suggests it too be reported. BFlh kinetic magnitudes peaked during HSR (8.95 m/s) at: force: 26.35 ± 5.15 N/kg, power absorption: −8.31 ± 3.34 W/kg, power generation: 5.0 ± 0.84 W/kg, total negative work: −0.34 ± 0.12 J/kg, and total positive work: 0.29 ± 0.05 J/kg {Schache, 2012 #6925}.

Primary joint kinetic data (detailed in Appendix S5) were extracted for knee flexion (internal moment: $n=44$,[23,35,49,54,60,62,69,70,73,76,77,79–110] power: $n=16$,[18,50,71,79–82,84,95,99,102,103,107,108,111,112] work: $n=2$[102,109]); and for hip extension (internal moment: $n=47$,[18,23,33,35,37,47–49,54,62,73–77,80,82,83,85–93,95,96,98,99,102–111,113–119] power: $n=12$,[33,71,82,95,99,102,103,107,108,111,112,116] work: $n=2$[102,109] and impulse: $n=1$[48]). Meta-analysis of sagittal plane kinetics was performed with eligible studies for knee flexion internal moment: fast (*swing phase only) (Figure 3, $n=5$, pooled mean: 2.2 Nm/kg, 95% CI: 1.9–2.5), slow (Figure 4, $n=4$, pooled mean: 0.9 Nm/kg, 95% CI: 0.4–1.4); for knee flexion power: fast (*swing phase only) (Figure 5, $n=5$, pooled mean: 40.3 W/kg, 95% CI: 31.4–49.2); for hip extension internal moment: fast (*swing phase only) (Figure 6, $n=4$, pooled mean: 4.8 Nm/kg, 95% CI: 3.9–5.6), slow (Figure 7, $n=7$, pooled mean: 2.3 Nm/kg, 95% CI: 1.9–2.7); and for hip extension power: fast (*swing phase only) (Figure 8, $n=3$, pooled mean: 33.1 W/kg, 95% CI: 17.4–48.9).

Data were also extracted for secondary vectors (Appendix S6) of knee IR (internal moment: $n=5$[69,83,93,98,102]), knee ER (internal moment: $n=15$,[50,52,58–61,66,69,83,86,88,93,102,120,121] impulse: $n=2$[52,121]), knee adduction (internal moment: $n=6$,[63,69,83,102,120,121] impulse: $n=1$[121]), knee abduction (internal moment: $n=27$,[44,45,50–54,56,58–61,63,66–70,75,83,86,88,93,97,98,102,111,121] impulse: $n=7$[51,52,56,57,67,122]), hip adduction (internal moment: $n=8$[53,55,75,83,88,89,102,117]), hip IR (internal moment: $n=8$,[48,50,83,89,93,98,102,117] impulse: $n=1$[48]), and hip ER (internal moment: $n=6$[61,83,89,93,102,117]). Meta-analysis of non-sagittal plane kinetics was performed with eligible studies for knee ER moment: slow (Figure 9, $n=3$, pooled mean: 0.20 Nm/kg, 95% CI: 0.04–0.37); and knee abduction moment: slow (Figure 10, $n=8$, pooled mean: 1.00 Nm/kg, 95% CI: 0.75–1.24).

4 | DISCUSSION

4.1 | Summary of findings

The present study aimed to summarize the available literature describing the activation and kinetic demands on the hamstrings in healthy individuals during high-speed running. Meta-analyses revealed highest muscle activation and kinetic demands at speeds approaching maximum. We were unable to perform a significant number of intended analyses in this review due to a paucity of research, heterogeneous data presentation, and inadequate methodological quality. No data were found on muscle function magnetic resonance imaging measures of muscle activation nor of rate of moment development in healthy individuals during high-speed running.

4.1.1 | Electromyographic studies

Of the 12 EMG studies included in the review, five normalized their magnitude data to activation levels present during maximal sprint speed. This procedure creates a ceiling of 100% activation at maximal sprint speed. We envisaged that the inclusion of these studies would help to determine the relationship between submaximal run intensities and their EMG magnitudes. If the notion that hamstring activation demands remain low until relatively high-run speeds, an early return to low-intensity running during rehabilitation could be justified. This

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Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Alt 2021b	2.65	0.11	0.01	2.43	2.87	23.54	0.00
Higashihara 2018	2.53	0.08	0.01	2.37	2.69	30.41	0.00
Nagahara 2017a	2.06	0.05	0.00	1.97	2.15	44.60	0.00
Schache 2011	1.76	0.10	0.01	1.57	1.95	17.78	0.00
Pooled	2.25	0.18	0.03	1.90	2.60	12.54	0.00
Prediction Interval	2.25			0.57	3.93		

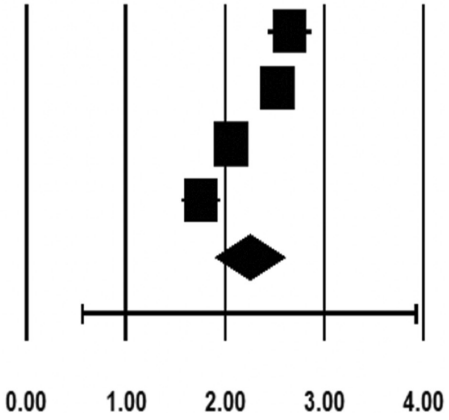

FIGURE 3 Forest plot of knee flexion swing phase moment (Nm/kg) during "fast" running (8.5+ m/s). Random effects model, $N=4$, heterogeneity: $I^2=95.0$.[23,80,99,102] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Kulmala 2013	0.90	0.16	0.03	0.59	1.21	5.60	0.00
Schache 2011	0.71	0.05	0.00	0.62	0.80	15.45	0.00
WIller 2021	1.50	0.05	0.00	1.40	1.60	28.93	0.00
Zhang 2021	0.50	0.04	0.00	0.41	0.59	11.16	0.00
Pooled	0.90	0.25	0.06	0.42	1.39	3.63	0.00
Prediction Interval	0.90			-1.46	3.26		

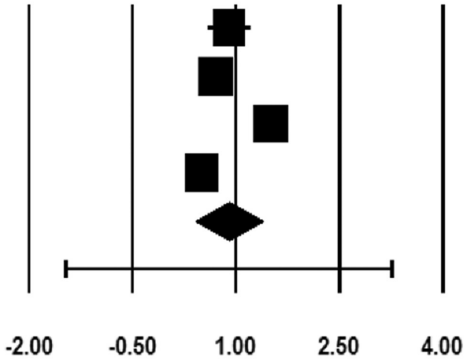

FIGURE 4 Forest plot of knee flexion moment (Nm/kg) during "slow" running (4.0–6.49 m/s). Random effects model, $N=4$, heterogeneity: $I^2=98.7$.[54,70,102,109] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

notion could not be confirmed or refuted through meta-analysis due to heterogeneous data presentation and low methodological quality within studies in this review. Examination of the five studies individually showed varied patterns of muscle activation in relation to run speed, with some reporting a nonlinear relationship (e.g., 70% maximal run speed ≈30% maximal medial hamstring activation)[43] and others a relatively linear relationship (e.g., 66% maximal run speed ≈66% maximal lateral hamstring activation).[123] Therefore, ascertaining conclusions on the relationship between submaximal run intensities and hamstring activation levels cannot be rendered from these studies.

4.1.2 | Kinetic studies

This review confirms that the hamstrings have a kinetically demanding task when approaching maximal sprinting and their primary function of hip extension and knee flexion exhibits increasing loads as speed increases. Some of the studies captured in the review of athletes running

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Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Alt 2021a	51.20	2.77	7.68	45.77	56.63	18.48	0.00
Alt 2021b	50.80	2.45	6.01	45.99	55.61	20.72	0.00
Haneda 2002	26.72	1.09	1.20	24.58	28.86	24.44	0.00
Nagahara 2017a	36.45	1.24	1.54	34.02	38.88	29.36	0.00
Schache 2011	37.15	2.55	6.48	32.16	42.14	14.59	0.00
Pooled	40.27	4.55	20.67	31.36	49.18	8.86	0.00
Prediction Interval	40.27			5.48	75.06		

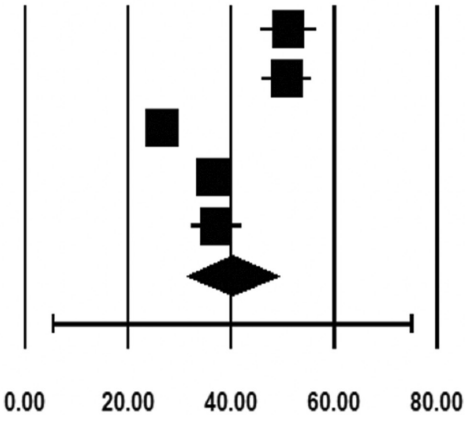

FIGURE 5 Forest plot of knee flexion swing phase power (W/kg) during "fast" running (8.5 + m/s). Random effects model, $N = 5$, heterogeneity: $I^2 = 97.0$.[71,79,80,99,102] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Alt 2021b	6.13	0.28	0.08	5.59	6.67	22.29	0.00
Higashihara 2018	4.44	0.24	0.06	3.97	4.91	18.40	0.00
Nagahara 2017a	4.25	0.16	0.03	3.93	4.57	25.83	0.00
Schache 2011	4.18	0.45	0.20	3.31	5.05	9.38	0.00
Pooled	4.76	0.45	0.20	3.88	5.64	10.59	0.00
Prediction Interval	4.76			0.62	8.90		

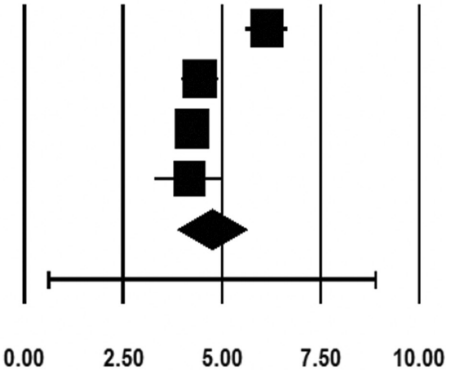

FIGURE 6 Forest plot of hip extension swing phase moment (Nm/kg) during "fast" running (8.5+ m/s). Random effects model, $N = 4$, heterogeneity: $I^2 = 91.8$.[23,80,99,102] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

at the fastest speeds reported knee flexion joint moments as high as 2.9 Nm/kg running at 9.8 m/s[79] and hip extension moments of 8.2 Nm/kg running at 9.7 m/s.[90] These extreme joint moment estimates should be interpreted with caution, however, as the methods used in the studies may overestimate joint kinetic measures when compared with subject-specific modeling.[124] Despite the use of similar methods in the studies captured in our meta-analyses, the estimates of joint moment demands during high-speed running (at 8.5 m/s or greater) produced a more conservative 2.2 and 4.8 Nm/kg joint moment for knee flexion and hip extension, respectively. Highest peak joint power measures were reported at high running speeds with pooled estimates for knee flexion and hip extension during the swing phase of fast running of 40.27 and 33.13 W/kg, respectively.

Point estimates of non-sagittal plane joint moments were far smaller than those of the primary outcomes, being 0.2 Nm/kg for knee ER and 1.0 Nm/kg for knee abduction joint moments. Although not captured in the review, coronal and axial plane joint kinetics may be of greater consequence to hamstring demands during

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Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Fukuchi 2015	2.55	0.19	0.04	2.17	2.93	13.32	0.00
Kulmala 2013	1.49	0.12	0.01	1.26	1.72	12.73	0.00
Richert 2019	1.90	0.13	0.02	1.65	2.15	14.72	0.00
Sanno 2018	1.83	0.12	0.01	1.60	2.06	15.85	0.00
Schache 2011	2.95	0.38	0.15	2.20	3.70	7.73	0.00
Stearne 2014 (FFS)	2.31	0.15	0.02	2.02	2.60	15.56	0.00
Stearne 2014 (RFS)	2.67	0.06	0.00	2.56	2.78	47.20	0.00
Willer 2021	2.87	0.09	0.01	2.69	3.05	32.04	0.00
Pooled	2.30	0.19	0.04	1.92	2.67	11.94	0.00
Prediction Interval	2.30			0.94	3.65		

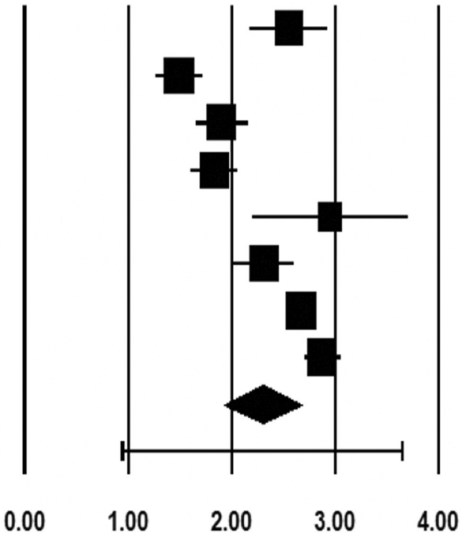

FIGURE 7 Forest plot of hip extension moment (Nm/kg) during "slow" running (4.0–6.49 m/s). Random effects model, $N = 8$, heterogeneity: $I^2 = 95.43$.[50,54,62,102,109,111,118] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Haneda 2002	19.63	1.09	1.20	17.49	21.77	17.95	0.00
Nagahara 2017a	39.25	2.55	6.48	34.26	44.24	15.42	0.00
Schache 2011	41.06	3.33	11.09	34.53	47.59	12.33	0.00
Pooled	33.13	8.05	64.79	17.35	48.90	4.12	0.00
Prediction Interval	33.13			-168.97	235.23		

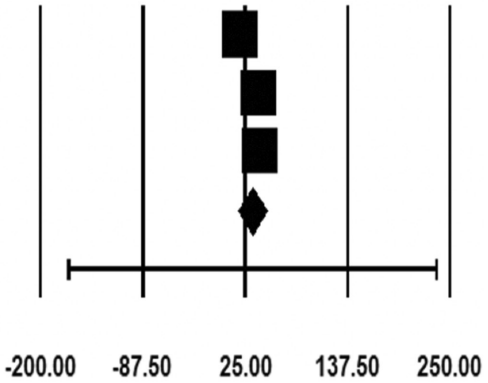

FIGURE 8 Forest plot of hip extension swing phase power (W/kg) during "fast" running (8.5+ m/s). Random effects model, $N = 3$, heterogeneity: $I^2 = 97.4$.[71,99,102] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

acceleration or change of direction tasks. This may be of relevance as recent studies on video analysis of hamstring injury mechanisms in rugby[125] and football[126] describe trunk rotation and change of direction motion of the player at the time of injury and the joint kinetics pertaining to the hamstrings in these tasks as warranting further investigation.

4.2 | Clinical implications

As high-speed running is reported as a common mechanism for HSI, understanding the demands of this potentially injurious task may help athletes, coaches, and performance staff to better match athletic capacity with task demands in an effort to reduce injury risk. The

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Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Fukuchi 2015	0.20	0.06	0.00	0.09	0.31	3.61	0.00
Schache 2011	0.32	0.03	0.00	0.26	0.38	10.06	0.00
Sinclair 2017	0.09	0.02	0.00	0.05	0.13	5.03	0.00
Pooled	0.20	0.08	0.01	0.04	0.37	2.43	0.02
Prediction Interval	0.20			-1.86	2.26		

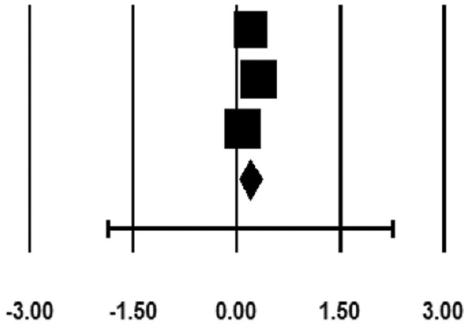

FIGURE 9 Forest plot of knee ER moment (Nm/kg) during "slow" running (4.0–6.49 m/s). Random effects model, $N=3$, heterogeneity: $I^2=95.1$.[50,66,102] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

Study name	Mean	Standard error	Variance	Lower limit	Upper limit	Z-Value	p-Value
Almonroeder 2015	0.54	0.06	0.00	0.42	0.66	9.11	0.00
Almonroeder 2016	0.70	0.05	0.00	0.59	0.81	12.99	0.00
Fukuchi 2015	1.18	0.14	0.02	0.91	1.45	8.68	0.00
Kulmala 2013	1.49	0.12	0.01	1.26	1.72	12.73	0.00
Schache 2011	0.99	0.17	0.03	0.65	1.33	5.71	0.00
Sinclair 2017	1.04	0.10	0.01	0.84	1.24	10.11	0.00
Stearne 2014 (FFS)	1.03	0.13	0.02	0.78	1.28	8.09	0.00
Stearne 2014 (RFS)	2.11	0.32	0.10	1.48	2.74	6.56	0.00
Zhang 2021	0.51	0.06	0.00	0.39	0.63	8.01	0.00
Pooled	1.00	0.12	0.01	0.77	1.23	8.53	0.00
Prediction Interval	1.00			0.18	1.82		

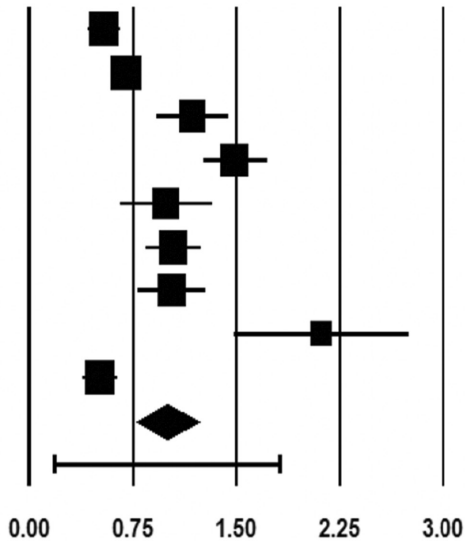

FIGURE 10 Forest plot of knee abduction moment (Nm/kg) during "slow" running (4.0–6.49 m/s). Random effects model, $N=9$, heterogeneity: $I^2=92.7$.[44,45,50,54,66,70,102,111] A prediction interval is an estimated range of values that are statistically predicted to contain the value of a single new observation. Contrary to confidence intervals, which focus on past or present event means, prediction intervals display reduced certainty, due to their focus on future individual events.

clinical implications of the current systematic review of hamstring demands in HSR are discussed below.

4.2.1 | Electromyographic studies

In the studies where data were normalized to MVIC, a notable finding was that activation consistently exceeded MVIC during maximal sprinting in either the medial or lateral hamstrings. In several cases,[40,41] even submaximal run intensities generated activation >100% MVIC. The highest activation level reported during maximal sprinting was 145% of MVIC for the lateral hamstrings.[34] The only EMG-based meta-analysis possible within this review was that of lateral hamstring activation during running at 90%–100% maximum intensity (Figure 2), which found a point estimate of 108% of MVIC. While movement artifact is an important consideration interpreting EMG data collected during high

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speed motions,[127] consistent "supra-maximal" activation levels does question the appropriateness of the selected isometric contractions in normalization during sprinting tasks. Suydam and colleagues[128] examined several "ballistic" EMG normalization strategies and compared them with MVIC. The authors found consistently higher and more reliable (via intra-class correlation coefficients) activation signals across tested muscles in sprinting compared with MVIC, thus highlighting the potential for a reevaluation of EMG normalization procedures.

Higher signal magnitudes in dynamic or ballistic contractions may relate to the need for explosive force, involving a rapid "burst" of motor unit recruitment, in order to initiate the muscle contraction[129] and force required. This may explain the hamstring EMG output disparity described by van den Tillaar and colleagues[130] between two "maximal" effort tasks, namely, the Nordic hamstring exercise (slow) and sprinting (fast). As the Nordic hamstring exercise intensity is contingent on slowing the rate of fall of the body due to gravity, there is typically a gradual increase in contraction intensity as the trunk segment lever arm extends in front of the pivoting knee as the body lowers. In contrast, maximal effort sprinting elicits both high force and rate of force development from the hamstrings, demonstrating a 30% or greater increase in EMG amplitudes.[130] If tasks requiring both maximal force and rate of force development elicit larger and more consistent signals from the nervous system to achieve the task demands,[128] then such tasks may provide more valid exercises for the optimization of athletic preparation and rehabilitation.

4.2.2 | Kinetics studies

The identification and assessment of functional markers of recovery are reported to be cornerstones of sports injury rehabilitation, as they facilitate a confident and timely return to training and competition.[131] A systematic review of return to play criteria following rehabilitation from HSI reports an over reliance on the alleviation of pain and an underutilization of objective measures to guide rehabilitation progression.[132] Data on kinetic demand retrieved by the present review may be applied in the rehabilitation setting to guide progression to return to play, where return to high-speed running is considered critical for performance but potentially risks re-injury. Clinicians may now add these objective criteria to manage re-injury risk as athletes progress through stages of running speed during rehabilitation, for example, have athletes generate >0.9 Nm/kg knee flexion before running at >4.0 m/s, or >4.8 Nm/kg hip extension before running >8.5 m/s.

Although further research could strengthen the certainty of the relationships presented, regression analysis between running speed and knee flexion (Figure 11) and hip extension (Figure 12) joint moments may serve as a tool for clinicians to estimate expected joint moments at a given speed when running. The large variation in methodology between studies and the small number of studies precluded running meta-analyses across running speeds. However, Figures 11 and 12 demonstrate a potential association between running speed and both knee and hip joint moments. Caution should be used when interpreting

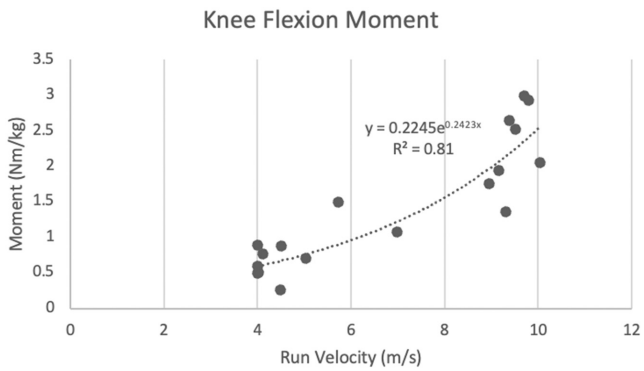

FIGURE 11 Knee flexion moment (*y* axis) graphed against run speed (*x* axis). Data were included from studies where data were normalized via Nm/kg and methodological quality scores were "moderate" or higher[23,49,50,54,55,62,70,79,80,90,95,99,100,102,107,109,111,115,117,118,138,139] The regression presented is only applicable for run velocities between 4 and 10 m/s. An exponential curve fitted to the data reports an R^2 value of 0.81, meaning 81% of the variance in knee flexion moment is explained by run speed.

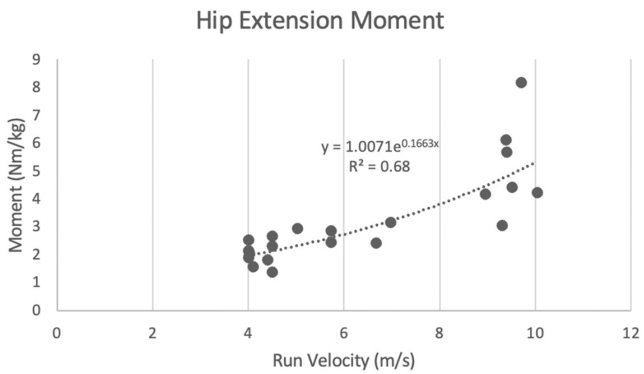

FIGURE 12 Hip extension moment (*y* axis) graphed against run speed (*x* axis). Data were included from studies where data were normalized via Nm/kg and methodological quality scores were "moderate" or higher.[23,49,50,54,62,80,90,95,99,102,107,109,111,115,117,118,138] The regression presented is only applicable for run velocities between 4 and 10 m/s. An exponential curve fitted to the data reports an R^2 value of 0.68, meaning 68% of the variance in hip extension moment is explained by run speed.

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these plots as the data used are for running velocities between 4 and 10 m/s. In addition, only the mean values for each study were used without accounting for sample size and within study variance. Future studies should evaluate such associations using meta-regression models. As research suggests that specific hamstrings have a bias toward either hip extension[133,134] or knee flexion[133,135] joint moment production, clinicians may further the specificity of testing in return to play decision-making by considering the specific muscle injured and weighting the test (either knee flexion or hip extension) appropriately.

4.2.3 | Research implications

As demonstrated by the low number of studies considered eligible for meta-analyses, the review highlighted a paucity of high-quality studies investigating the activation and kinetic demands on the hamstrings during HSR. Further research using consistent methods will enhance understanding of hamstring muscle demands in HSR by allowing more data pooling in meta-analyses.

The strong relationship between running speed and joint moments observed across studies here (Figures 11 and 12) is consistent with the expected increase in kinetic demands on the hip and knee joints as speed increases. One study with a unique perspective,[116] however, did not support this relationship. This inconsistency was seen in data from O'Meara and colleagues,[116] who reported hip extension moments across different run speeds in three distinct cohorts of athletes; sprinters (mean speed: 9.3 m/s), middle-distance runners (mean speed: 7.5 m/s), and triathletes (mean speed: 5.5 m/s). Despite demonstrating the highest mean run speeds, sprinters displayed similar peak hip extension moments (3.5 Nm/kg) to the middle-distance runners (3.5 Nm/kg) and triathletes (3.7 Nm/kg). One explanation for this could relate to differences in running mechanics, as athletes can display individual strategies to execute complex tasks, influenced by factors such as training background and implicit performance attributes. The nature of the relationship between HSR strategy and the kinetic demands on the hamstrings at a given speed is yet to be established, as most studies investigating HSI have been based on kinematic rather than kinetic variables.[19] Hence, further research on this relationship is warranted.

While both muscle functional magnetic resonance imaging and rate of moment development measures of hamstring demands were sought out in the review search strategy, neither metric was found in any relevant HSR literature. Future studies assessing muscle functional magnetic resonance imaging before and after HSR in healthy subjects could therefore help better define the spatial activation patterns of the task to ascertain the superficial and deep muscles contribution to HSR. Similarly, future studies examining rate of moment development for hamstring-related vectors during HSR (primarily knee flexion and hip extension) could elucidate the explosive nature of hamstring demands in this challenging task. Knowledge of these measures could enhance the design of athlete injury resilience strategies through greater specificity in training plans and more objective markers of rehabilitation progression criteria.

This review focused only on steady state, straight line HSR in healthy subjects. Studies of subjects with previous HSI were excluded and the kinetics of these individuals during HSR may differ from healthy participants. Although HSR is implicated in the majority of proposed HSI mechanisms,[11] the results of the present study cannot be fully representative of the demands present in sports-based HSI. Alongside investigating previously injured participants, future reviews should quantify hamstring activation and related kinetics during the acceleration phase of running, high-speed curvilinear running and changes of direction.

4.2.4 | Limitations

The database search was limited to articles published in English. Relevant studies may be published in non-English language journals, but with a total 96 included articles, well above average for systematic reviews,[136] it was considered that the selection made would be representative of the broader literature.

Low-methodological quality prevented inclusion in meta-analysis of many studies, typically due to dated or suboptimal methods in data acquisition (e.g., forward dynamics with static optimization for kinetic variables, low sampling rates of equipment) which display inferior accuracy when compared with more contemporary approaches (i.e., subject-specific modeling).[124] All meta-analyses conducted reported high heterogeneity ($I^2 > 90\%$). Meta-regression and subgroup analyses to elucidate potential sources for this were not possible due to the low number of eligible studies reporting on each variable. More consistent methods in data acquisition and presentation will enhance the value of future reviews and the certainty of inferences made from meta-analysis.

A significant portion of the review focused on kinetics pertaining to the hamstring muscles. While articles calculating muscle kinetics may be considered specific to the hamstrings, joint-based kinetic metrics may not. The first reason for this is that joint moments are a "net" calculation using inverse dynamics calculation, meaning they

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can only inform on the instantaneous *balance* of forces (i.e., a magnitude in favor of knee flexion over extension, not absolute values of each). Hence, joint moments do not account for antagonist muscle contribution to instantaneous joint moment. Although the concept of reciprocal inhibition suggests that agonist muscle contraction stimulates antagonist muscle inhibition,[137] any co-contraction from antagonist muscle would lead to underreporting of demands on the hamstring muscles. The second reason is that the hamstrings are not the only muscles responsible for motion of the joints investigated. For example, a significant portion of hip extension moments may be explained by gluteal muscle contraction, though hamstrings likely remain responsible for most of the internal knee flexion moment.[77] However, if clinical testing replicates the joint positions where peak moments occur in sprinting, then hip extension moments involving all hip extensors will maintain ecological validity for measures captured during HSR.

Finally, while we advocate for the value that our metrics for hamstring activation and related kinetic variables can add to HSI prevention and rehabilitation, we do not imply that simple isometric clinical testing replicates the exact conditions of sport-induced HSI. The data presented may inform only a component of the complex processes involved in the design of injury prevention program and return to play decision-making strategies.

5 | PERSPECTIVE

High-speed running is a common mechanism for HSI,[10,11] yet activation and kinetic demands on the hamstrings are not well defined. The current review summarizes the known literature regarding hamstring activation and related kinetic demands during high-speed running. Performance and rehabilitation clinicians may use the presented data to design return to run and injury prevention/rehabilitation plans informed by objective criteria.

6 | CONCLUSION

Faster running intensity/speed places higher activation and higher kinetic demands on the hamstring muscles. Joint moments to which the hamstrings contribute also increase with increasing running speed, although some intersubject variability exists. While concerns about reporting and methodological quality prevent firm conclusions regarding the magnitude of hamstring demands, clinicians may still find the data presented here to be of value in training, testing, and return to play decision-making strategies. Future research on activation (particularly muscle functional magnetic resonance imaging) and kinetic (particularly rate of moment development) measures is warranted to further enhance our understanding of the demands on hamstring muscles of healthy individuals during HSR.

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CONFLICT OF INTEREST STATEMENT

All authors declare that they have no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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